

Behavioral Finance Analysis Using AI on Player Spending Patterns in GTA V

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Background & Purpose

Introduction

GTA V — with **185M+ copies sold** — features a dynamic virtual economy where players manage risk, allocate resources, and make spending decisions mirroring real-world finance.

Classical behavioral finance (Kahneman & Tversky, 1979) shows decisions are driven by emotion and cognitive bias, not pure rationality. Virtual economies offer a rich, data-rich environment to study these patterns using AI.

- ❏ This study bridges behavioral finance theory and AI-driven game analytics to decode player financial psychology.

Research Gap & Prior Work

Literature Review

20 peer-reviewed papers reviewed. Key gap: no prior study combines LSTM, Random Forest, and NLP together to analyze virtual spending behavior holistically.

Behavioral Finance Foundations

Prospect Theory (Kahneman & Tversky, 1979): loss aversion 2× stronger than gain. Mental accounting explains why Shark Card money is spent more freely than earned GTA\$.

AI in Game Analytics

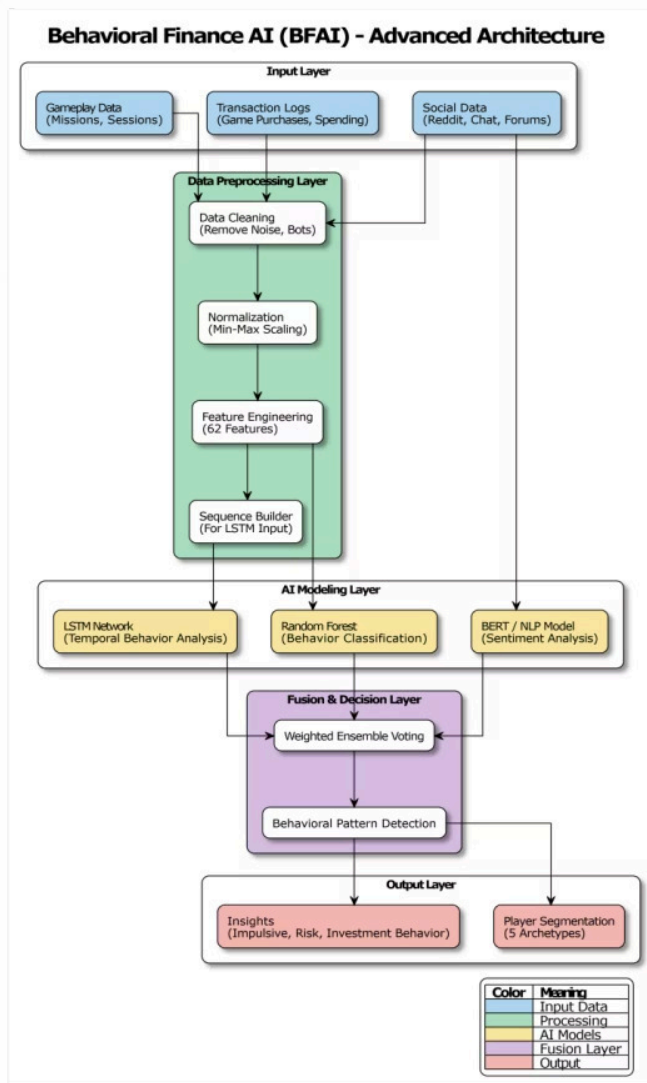
Mehta et al. (2022): RNN achieves 85.2% accuracy predicting high-value spenders. Sharma & Gupta (2021): Gradient Boosting F1=0.83 for impulsive spender classification.

Impulsive & Risk Behavior

Reddy & Krishnamurthy (2023): scarcity and social comparison drive impulsive microtransactions. Nair & Balachandran (2022): 71% of GTA players experienced virtual buyer's remorse.

Virtual Investment Behavior

Desai & Iyer (2021): herding and disposition effect observed in GTA's stock market. Patel & Shah (2020): $r=0.67$ correlation between in-game risk and real-world risk tolerance.



System Architecture Diagram





Research Goals

Objectives

01

Build the BFAI Framework

Develop an AI system to detect and classify behavioral finance signals — risk-taking, impulsive buying, and investment-like decisions — from GTA V game data.

02

Measure Behavioral Frequencies

Quantify how often players exhibit risk-seeking, impulsive spending, and investor-like asset allocation strategies.

03

Validate Against Theory

Compare in-game behavioral patterns to established real-world behavioral finance concepts (loss aversion, herding, endowment effect).

04

Derive Practical Implications

Inform game design, player welfare policies, and digital economy regulation based on findings.

Methodology



Trained on 80% of 243,780 transaction logs from 5,614 players (Jan-Dec 2023). NLP applied to 18,340 Reddit posts and 94,500 comments from r/gtaonline.

LSTM

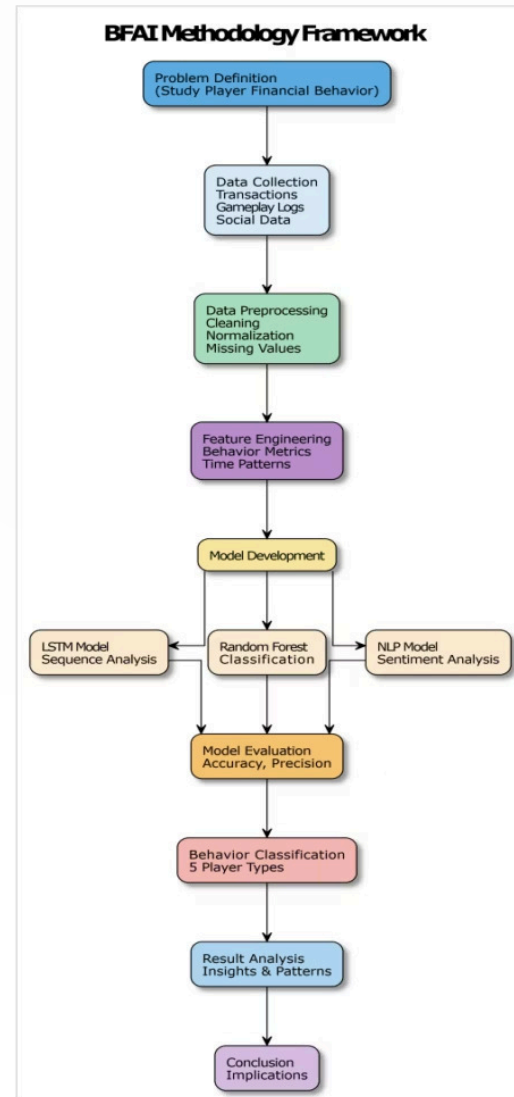
3-layer, 256 units. Dropout 0.3. 100 epochs, Adam optimizer. Trained on NVIDIA A100.

Random Forest

500 trees, depth 20. Top features: Shark Card frequency, mission risk score, asset-to-spending ratio.

NLP Pipeline

BERT fine-tuned with FinBERT weights. Sentiment scored -1.0 (fearful) to +1.0 (optimistic) weekly.



Methodology Diagram

Results & Findings

89.4%

Classification Accuracy

Overall BFAI framework accuracy across 5 behavioral archetypes.

43.1%

Investor-Like

Players demonstrating portfolio diversification mirroring real-world behavior.



Impulsive Spenders · 34.2%

Strategic Investors · 21.7%

Risk-Seekers · 18.4%

Loss-Averse Hoarders · 15.3%

Social Herders · 10.4%

67.3%

Impulsive Buyers

Players showing event-proximity buying within 15 min of in-game events.

91.2%

Risk-Seeker Detection

LSTM accuracy identifying Risk-Seeker archetype session patterns.

Behavioral Finance Parallels

- **Loss Aversion:** Players refuse to sell depreciated vehicles — detected via LSTM holding-period sequences.
- **Mental Accounting:** Shark Card money spent more freely than earned GTA\$ — top RF feature.
- **Herding:** Mass DLC purchases correlated with NLP community trend spikes.
- **Strategic Investors:** 84.3% stock market participation; forum sentiment correlated with investment activity ($r=0.58$, $p<0.01$).



Summary of Contributions

Conclusion



For Game Developers

Use behavioral archetypes to design safeguards protecting players prone to excessive spending.



For Regulators

Virtual economies warrant consumer protections similar to real financial markets.



For Researchers

GTA V is a valid, data-rich laboratory for studying real-world financial decision-making under risk.

The BFAI framework successfully classified players into 5 archetypes with **89.4% accuracy**, confirming that virtual spending behavior mirrors classical behavioral finance theory — including loss aversion, mental accounting, herding, and the disposition effect.

Areas for Further Research

Future Scope

→ Cross-Game Validation

Extend the BFAI framework to other virtual economies (e.g., Fortnite, Roblox) to test generalizability.

→ Longitudinal Behavior Tracking

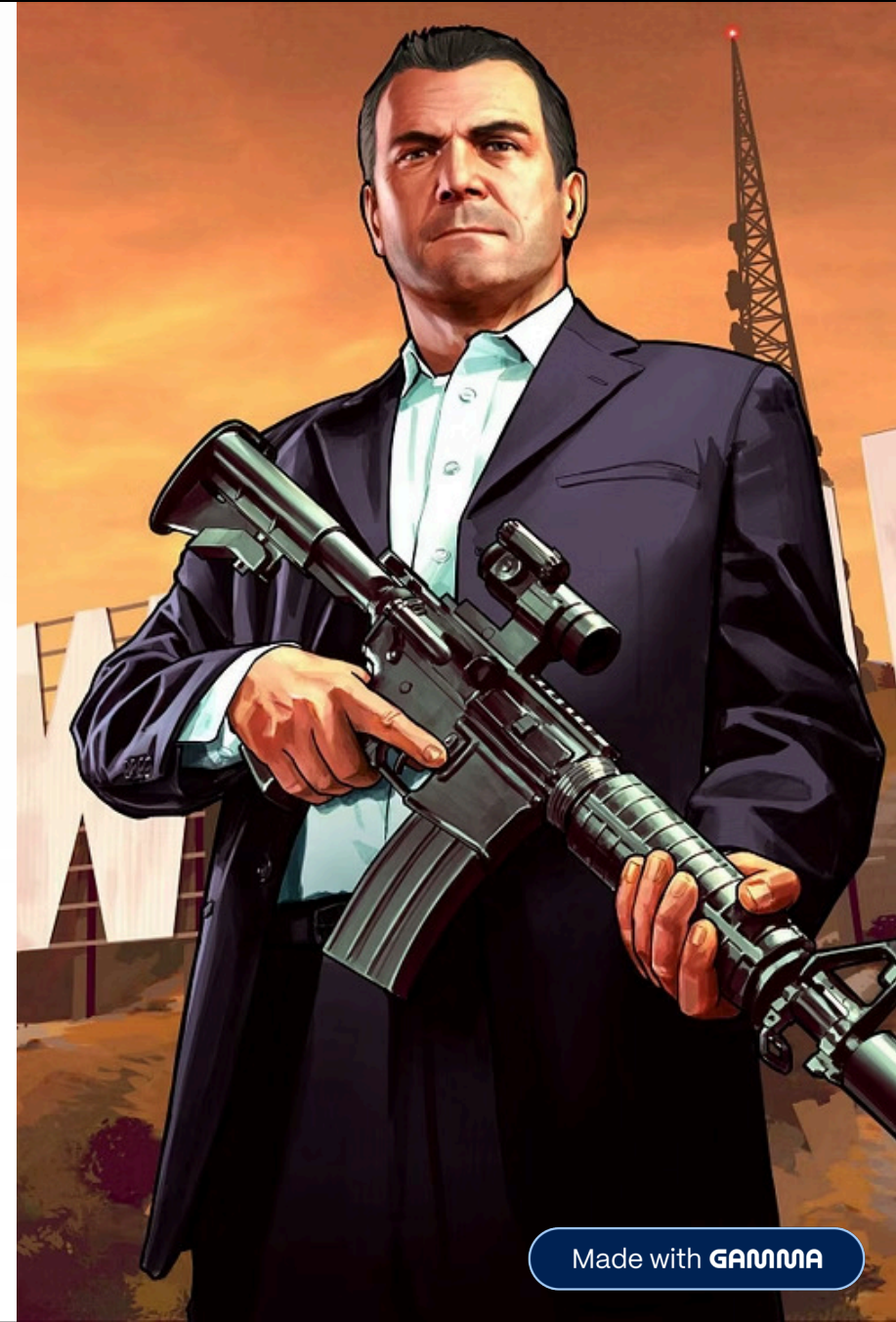
Study how player archetypes evolve over time as game economies and updates change.

→ Real-World Correlation Studies

Directly link in-game financial behavior to participants' real-world financial decisions and risk tolerance.

→ Regional & Cultural Analysis

Examine behavioral differences across South Asian and global player populations for culturally inclusive insights.



References

Behavioral Finance Theory

Kahneman & Tversky (1979) — Prospect Theory. Thaler (1985) — Mental Accounting. Loewenstein et al. (2001) — Risk as Feelings. Shiller (2019) — Narrative Economics.

Global AI & Consumer Research

Yannakakis & Togelius (2018) — AI and Games. Johnson et al. (2020) — Emotion from gameplay. Rook & Fisher (1995) — Impulsive buying. Weber et al. (2002) — Risk attitudes. Lehdonvirta (2009) — Virtual goods. Hamari et al. (2017) — Gamification & spending loops.

Indian Gaming Studies

Sharma & Gupta (2021); Mehta et al. (2022); Reddy & Krishnamurthy (2023); Nair & Balachandran (2022); Patel & Shah (2020); Desai & Iyer (2021); Joshi & Pillai (2022); Singh & Verma (2023); Krishnan & Bose (2024); Bhatt & Agarwal (2023).

❏ Full citations with DOIs available in the published paper. All 20 references are peer-reviewed and directly inform the BFAI framework's theoretical and methodological foundations.